

Challenge (Habanero)

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + 2y \leq 6$ and $2x + y \geq 4$ on the same coordinate plane. Identify the overlapping solution region and interpret the feasible region in the context of a construction project where x represents the number of workers and y represents the number of machines.
- Q10.** A builder is designing a new house with a budget of 250,000. The cost of the foundation is 40,000 and the cost of the roof is 30,000. If the cost of the walls is x and the cost of the windows is y , where $x + 2y \leq 120,000$ and $2x + y \geq 60,000$, determine the maximum amount of money the builder can spend on the walls and windows.

Chef's Tips

- Ensure students have graph paper for questions 9-10
- Remind students to show all work and explain their answers for questions 9-10
- Encourage students to use different colors to graph the inequalities in questions 9-10